

Chapter 11

The Mechanisms of Emergence



Stuart Glennan

Abstract Emergentism is often imagined to be opposed to mechanism. If some phenomenon admits of mechanistic explanation, it is thought to be *ipso facto* not emergent. In this paper I argue to the contrary that emergence requires mechanism. Whenever some emergent phenomenon occurs, there is a mechanism responsible for its emergence. To make this case I show how mechanisms can explain four commonly held characteristics of emergent phenomena – dependence, autonomy, novelty and holism. By looking at the various kinds of emergence-generating mechanisms, it will be possible to classify different kinds of emergent phenomena by the particular features of the mechanisms that generate them, and so to bring some order to diversity of phenomena that we call emergent.

Keywords Emergence · New mechanism · Dependence · Autonomy · Novelty · Holism · William Wimsatt

11.1 Introduction: Mechanisms and Emergence

While there is no consensus on what emergence is or where and when it occurs, there is widespread agreement that it is opposed to mechanism. That opposition dates back to the foundations of the contemporary emergence debate in C.D. Broad's *Mind and its Place in Nature* (1925). Relatedly, whatever emergence is, it is generally understood to be opposed to reductionism. A reductive explanation of some phenomenon is taken to show that the phenomenon is not emergent.

In this paper I argue that these assumptions about emergence are misguided. Any emergent phenomenon must emerge *from somewhere*, and this somewhere is the mechanism that is responsible for the emergent phenomenon. I shall argue for this conclusion by showing how mechanisms can be responsible for phenomena that

S. Glennan (✉)
Butler University, Indianapolis, IN, USA
e-mail: sglennan@butler.edu

exemplify widely agreed upon criteria of emergence. Moreover, different kinds of mechanisms will satisfy these criteria in different ways and to different degrees, suggesting we can use kinds of emergence-generating mechanisms to distinguish different kinds and degrees of emergence.

The account I will offer here owes much to the work of Bill Wimsatt. Wimsatt's basic formulation of emergence is this:

Emergence of a system property relative to the properties of the parts of that system indicates its dependence on their mode of organization. It thus presupposes the system's decomposition into parts and their properties, and its dependence is explicated via a mechanistic explanation (Wimsatt, 2007, 276).

Wimsatt calls these emergent system properties “non-aggregative” because their dependence on mode of organization implies that you cannot aggregate the parts any which way and still recover the emergent properties. Purely aggregative properties – things like mass and charge – are rare, so emergence is the exception rather than the rule. Furthermore, Wimsatt sees no tension between emergence and reduction. For Wimsatt, *[a] reductive explanation of a behavior or a property of a system is one that shows it to be mechanistically explicable in terms of the properties of and interactions among the parts of the system* (ibid, 275; italics in original).

To defend a view like Wimsatt's, we must show how, despite the conventional wisdom about the opposition between emergence and mechanism, mechanisms can in fact explain how phenomena emerge. My strategy will be to identify four widely agreed upon principles for what is required for some phenomenon to be emergent, and then show how mechanisms can account for them.

The most commonly cited principles generally go by the names of dependence and autonomy (Gibb et al., 2019; O'Connor, 2021; Wilson, 2021). To these I add two additional criteria, that I call novelty and holism. These criteria are, on many taxonomies, aspects of the autonomy principle, but novelty, holism and autonomy can split apart and are sometimes in tension, so I will, following (Humphreys, 2016), keep all four. Here are one-sentence characterizations of each:

- **Dependence** — Emergence is a dependence relation between the source of emergence (the emergence base) and the result of the emergence (the emergent phenomena).
- **Autonomy** — Emergent phenomena should be autonomous from their emergence base.
- **Novelty** — Emergent phenomena have novel features that do not belong to the base from which they emerge.
- **Holism** — In emergent phenomena, the whole is more than the sum of the parts.¹

Here is the plan for the rest of the paper. In Sect. 11.2 I will provide a quick review of key features of mechanisms, as they have been articulated in recent discussions

¹Humphreys describes his four criteria as follows: “Emergent features result from something else, they possess a certain kind of novelty with respect to the features from which they develop, they are autonomous from the features from which they develop, and they exhibit a form of holism” (Humphreys, 2016, 26). I take his first criteria as expressing what I am calling “dependence.”

of mechanisms in the philosophy of science. I will use this account of mechanisms in Sect. 11.3 to interpret the dependence relation required for emergence as the relation of mechanism-dependence. I will clarify what can emerge from what by describing possible relata of this dependence relation, and I will use a basic distinction between two kinds of mechanisms to explicate the much-discussed distinction between synchronic and diachronic emergence. In Sect. 11.4, I will show how different kinds of mechanisms can generate phenomena that are in various ways autonomous, novel and holistic; I will also show how these three principles are related to some other features commonly held to be characteristic of emergent phenomena – for instance, downward causation, self-organization, and multiple realization. In the last section I will recap to what for some may be a nagging question – whether a mechanistic and reductionist theory of emergence misses the essence of emergence.

11.2 Mechanisms and their Varieties

The basic supposition of a mechanistic theory of emergence is that emergent phenomena emerge out of the activities of mechanisms, and that different varieties of emergent phenomena can be identified by comparing the kinds of mechanisms which generate them. In order to make a case for such a theory, I will begin with a review the basic features of mechanisms as articulated within the new mechanist literature in philosophy of science (Craver & Tabery, 2016; Glennan & Illari, 2018a; Glennan et al., 2021) and to describe briefly an approach to classifying varieties of mechanisms that I have developed elsewhere (Glennan, 2017).

In ordinary English usage, the word ‘mechanism’ has two senses. In the first, mechanisms are systems with interacting parts, often but not always artifacts; in the second, mechanisms are processes which bring about some happening or activity. Example of mechanisms in the first sense are things like clocks or computers; examples of mechanisms in the second sense are mechanisms of protein synthesis or reproductive mechanisms. Call these two senses respectively the systemic and the processual sense of mechanism.²

Within most scientific contexts, the processual sense of mechanism is the more common, and for this reason, new mechanistic approaches have taken the processual sense to be primary. The new mechanistic account of mechanisms can be briefly summarized in a definition I call minimal mechanism: “a mechanism for a

²The Oxford English dictionary lists these two definitions for mechanism:

1. a system of parts working together in a machine; a piece of machinery
2. a natural or established process by which something takes place or is brought about

These definitions suggest that the system sense is connected to machines and machinery, while the processual sense is connected with natural phenomena. While there is no doubt that many of our stereotypic examples of mechanistic systems are machines, there is nothing in the idea of a system that requires it to be of artificial construction; and similarly, there is nothing in the idea of a mechanistic process that requires it to be “natural.”

phenomenon consists of entities ...whose activities and interactions are organized so as to be responsible for the phenomenon” (Glennan, 2017; Glennan & Illari, 2018a)

According to minimal mechanism, mechanisms are individuated by what they do — their phenomena or behavior. One speaks of the mechanism of protein synthesis, or predation or reproductive mechanisms, or the mechanisms by which animals communicate, or by which national banks control the money supply. All of these are phenomena that depend upon mechanisms.

The mechanisms responsible for such phenomena are made up of constituents that are termed **entities**, **activities**, and **interactions**. Entities are understood to be “things” — objects, systems, structures, etc.³ They could be proteins, cells, organisms, families, baseballs, televisions, planets, stars or galaxies. The activities and interactions are the doings in which these entities engage — e.g., bonding, folding, striking, heating, walking, eating, radiating, exploding. Activities are processual in the sense that they are temporally extended, often but not always with distinct beginnings, intermediate stages and ends.

The difference between activities and interactions is simply the number of actors. Interaction requires multiple entities to be involved, whereas activities may involve only a single entity; for instance, sexual reproduction takes two actors, while asexual reproduction takes only one. Going forward, I will sometimes speak generically of activities as being inclusive of both one-place “solo” activities and multi-place interactions.

In order for a mechanism to give rise to some phenomena, its constituent entities, activities and interactions must be organized in a particular way. When, for instance, an animal turns its body, this activity (the mechanism’s phenomenon) requires that the parts of the animal (its joints, limbs, muscles, etc.) are put together in a certain way, and that the timing of the activities of and interactions between these parts are coordinated. The general lesson is that a pile of mechanism parts does not make a mechanism. This is the minimal sense in which mechanisms are always more than the sum of their parts.

It is helpful to think about mechanisms as composite processes. They are processes, because mechanistic phenomena always involve activities and interactions, which are temporally extended doings. They are composites, because these doings depend upon a mechanism constituted by its components — organized activities and interactions of some set of underlying entities; these components are said to be **constitutively relevant** to the mechanism.

Take, for instance, protein synthesis. This is an activity or process, and it is constituted by the activities and interactions (transcription, translation, etc.) of the various entities (DNA, mRNA, etc.) that are collectively responsible for the activity of

³This use of the term ‘entity’, which originates with (Machamer et al., 2000), is unfortunate given the fact that metaphysicians often use ‘entity’ to refer generically to anything that belongs in one’s ontology — including objects, but also properties, relations, laws, tropes, and whatever. I will occasionally switch between the new mechanist’s and the metaphysician’s sense, trusting to the context to make the intended meaning clear.

protein synthesis. It is because mechanistic phenomena always involve activity that Craver et al. (2021) suggest that all mechanisms have a “processual core.”

Entities are also composites, but they are not mechanisms in the processual sense of minimal mechanism.⁴ Their components are other entities. For instance, a cell is an entity composed of its membranes, organelles, and so on. These composite entities are generally what I’ve called **mechanistic systems** — composite entities whose persistence and interactions with the world depend upon mechanisms constituted by the activities and interactions of their components. A cell, for instance, is a mechanistic system because it cannot live and perform its functions within its environment without the constant operation of mechanisms involving organized activities and interactions of its parts.

One of the chief payoffs of a mechanistic account of emergence is that it will allow us to classify different varieties of emergence as arising from different varieties of mechanisms. The terms within the minimal mechanism definition suggests four dimensions of classification (Glennan & Illari, 2018b):

- the kinds of phenomena for which the mechanism is responsible
- the kinds of entities the mechanism has as constituents
- the kinds of activities and interactions their constituent entities engage in
- the ways in which entities and activities/interactions are organized within the mechanism

A fifth dimension concerns not current features but history. Mechanisms may be classified etiologically, by how they came to be.

These five dimensions are largely independent, so, for instance, it is possible for mechanisms with very different kinds of constituent entities and activities/interactions to have similar kinds of organization. In what follows we shall see that we can understand different varieties of emergence chiefly in terms of different kinds of phenomena, different kinds of organization and different kinds of etiology.

11.3 Emergence as Mechanism-Dependence

Minimal mechanism holds that all mechanisms are mechanisms **for** some phenomenon, and any such phenomena can be said to be **mechanism-dependent**. A mechanistic account of emergence proceeds from the supposition that the dependence between emergent phenomena and their emergence bases are relations of mechanism-dependence. Mechanism-dependence is not sufficient for emergence, since by itself it does not guarantee autonomy, holism or novelty, but it is necessary. In this section I will show how recognizing different varieties of mechanism-dependence relations and different possible relations yields a natural way of describing different varieties of emergence.

⁴See (Glennan, 2021) for my preferred account of this composition relation.

11.3.1 *Producing versus Underlying and the Distinction between Diachronic and Synchronic Emergence*

Perhaps the most basic distinction to make between kinds of mechanisms is the distinction between mechanisms that produce phenomena and mechanisms that underlie phenomena. In the former case, there is a mechanistic process ψ that is triggered by some set of startup conditions ψ_{in} and terminates with a product ψ_{out} .⁵ For instance, in a protein synthesis mechanism, ψ_{in} would be the set of conditions initiating protein synthesis, and ψ_{out} would be the protein product. In the latter case, the phenomenon in question is the activity or process itself, and the mechanism underlying it is the set of organized activities and interactions of entities that constitute that activity. For instance, if a muscle (S) contracts (ψ) then what underlies this contraction (i.e., what constitutes it) are the contractions (ϕ_i) of the many muscle fibers (X_i) that make up the contracting muscle.

The distinction between producing and underlying relations corresponds to the familiar distinction between diachronic and synchronic emergence (see e.g., Humphreys, 2016, sec. 1.7.4). In diachronic emergence, we can interpret the emergence base as the set of startup conditions which, via an etiological mechanism, produces the emergent phenomenon. The emergence base is temporally prior to and distinct from the emergent phenomenon, and the etiological mechanism is the causal process by which the emergent phenomenon emerges. In synchronic emergence by contrast, the emergent phenomenon depends upon an underlying mechanism, which coexists with the phenomenon in space and time. This interpretation of synchronic emergence permits the relation between the emergence base and the emergent to be temporally extended and dynamic. If, for instance, a behavior of an animal emerges synchronically from the activities of the animal's parts, the behavior and the underlying mechanism will both involve temporally extended activities and interactions.

The same phenomena may emerge both diachronically from a temporally prior emergence base and synchronically from a temporally overlapping emergence base. An organism is a mechanistic system that emerges in both of these senses; it emerges diachronically from developmental mechanisms and synchronically from the entities and activities which underlie and maintain the organism and its activities. The two varieties of emergence correspond to the two aspects of mechanistic explanation. One explains diachronic emergence via an etiological mechanistic explanation, while one explains synchronic emergence via a constitutive mechanistic explanation.

⁵I make use here of Craver's (2007) formalism that denotes activities by Greek letters and entities by Roman letters. Since many mechanistic processes are cyclical (as for instance in metabolic pathways), the startup and termination points may be arbitrarily identified points within ongoing cycles.

11.3.2 *What Emerges: The Relata of Mechanism-Dependence Relations*

Since emergence is a kind of dependence relation, one way to distinguish its varieties is by the varieties of relata that may stand in this relation. In the metaphysics literature the relata are often assumed to be properties, but interpreting emergence as mechanism-dependence suggests that things other than properties can stand in emergence relations. We can sort emergent relata into the following categories:

- Emergent processes, activities and interactions
- Emergent entities or systems
- Emergent properties and relations

Each of these kinds of emergents may emerge both synchronically from an underlying mechanism or diachronically from an etiological mechanism.

Consider first activities. A mouse running through a maze is a standard example of what in the mechanisms literature is called an “entity acting” (Krickel, 2018). The mouse’s running ($S \psi$ -ing) depends upon the orchestrated activities and interactions of the entities that underlie this ψ -ing. These include activities of and interactions between elements of the muscular/skeletal system, the cardiovascular system and the central nervous system, among others. The relation between the mouse’s running and the activities and interactions of its parts is synchronic and constitutive. When the mouse moves a leg, that movement is not the cause of the mouse’s running but is part of the activity of its running.

While some activities are aptly described as entities acting, not all are. Many activities are in fact interactions between two or more entities (in which case we would say, e.g., that S and T are ψ -ing together). One example is the mechanism by which one or more neurons trigger another neuron across a synapse. In such a case there are constituent entities within S and T , as well as entities in the environment (e.g., neurotransmitters in the synapse) whose activities and interactions underlie the interaction between the triggering and triggered neurons.

The mouse running or the neuron triggering are plausible candidates for synchronically emergent activities, though to make the case fully we would have to consider how they might meet the autonomy, novelty and holism requirements which we will take up in the next section. It is less plausible to think that particular instances of such activities emerge diachronically. We give etiological mechanistic explanations of these processes, but they would largely be accounts of the events that produce the startup conditions for these mechanisms. For instance, we could give an etiological explanation that identified the stimulus that made the mouse start running. But such an explanation is straightforwardly causal, and something more than bare causal dependence is needed for emergence.

Better candidates for diachronic emergence of activities are atmospheric processes like winds, rains and hurricanes. Consider as an example the trade winds. The trade winds are relatively constant easterly winds that blow in the tropical

regions north and south of the equator. They are generated by an underlying mechanism involving both the cycling of air and moisture from the equator towards the poles and the earth's rotation. The conditions that give rise to these winds have been fairly constant throughout recorded human history, but they depend for their existence upon features of the planet and atmosphere that have changed over time. For instance, the current wind patterns depend upon the location of continental land masses, and those have shifted over time. Because these historical conditions generated processes that are novel and self-organizing, the resulting trade winds seem like plausible candidates for diachronically emergent processes.

The second category of emergents is entities/systems. In what sense might they depend upon mechanisms, and thereby emerge from the activities of mechanisms? Most obviously, entities can be the products of producing mechanisms: proteins are produced by a synthesis mechanism, cars are produced by an assembly line, and organisms are produced by reproductive and developmental processes. If these entities are indeed emergents, this variety of emergence would be diachronic. Such products depend diachronically upon their antecedents, and they "emerge" in some pre-theoretical sense as the product of the mechanism. Which such products should count as genuinely emergent will depend upon the degrees and respects in which the products meet the autonomy, novelty and holism criteria.

It is less obvious whether and when entities can rightly be said to emerge synchronically from constitutive mechanisms, but if the entity is what I've called above a mechanistic system, it is plausible to say that the system emerges from the activities of its constituents. Consider a mouse. While the mouse is not a mechanism in the processual sense of minimal mechanism, its persistence as a living mouse requires the action of many mechanisms involving its constituent entities and their actions and interactions, both among themselves and with their environment. A mouse is sustained for instance by its metabolism, by its moving about to find food and evade predators, and so forth. Since its continuation as a living mouse depends synchronically upon these mechanisms, there is a clear sense in which we can understand these mechanisms as an emergence base from which the living mouse synchronically emerges.

What of emergent properties and relations? Certainly there are such things, but a mechanistic theory of emergence gives a different account of their nature than most metaphysical accounts. Whereas abstract metaphysical accounts of emergence conceive of emergence relations as modal relations like supervenience or grounding, the mechanistic account understands properties to belong to composite entities. It takes a broadly causal/dispositional view of properties — a view that properties are individuated by their causal role. To say, for instance, that the table is solid, is to say that other objects will not fall through it; or to say that a piece of paper is flammable is to say that it will catch fire when touched by a flame. The mechanist's insight is simply to say that composite entities have such dispositions in virtue of being mechanistic systems — systems within which mechanistic processes involving their components can be triggered to manifest such dispositions. A paper is flammable because interactions like touching it with a lighted match will trigger its burning,

while a fireproof cinder block, because it is composed of different kinds of entities, differently organized, will react differently to a flame. If this view of properties is correct, then all but the most fundamental properties will be properties of composites, and their manifestation will depend upon mechanisms. Relations between properties and the mechanisms upon which they depend are synchronic, because properties manifest themselves through the activity of underlying mechanisms.

Properties also emerge diachronically from etiological mechanisms that produce changes in the properties of entities. Consider again the mouse and its running. The mouse's capacity to run is a behavioral property of the mouse — a disposition that manifests itself when some stimulus or cognition triggers it. This disposition is not something the mouse is born with; it emerges gradually as the mouse develops and interacts with its environment. More generally, when we consider the traits of biological organisms (anatomical, physiological, or behavioral), we will find that they emerge in individual organisms via developmental mechanisms of various kinds. A different kind of diachronic emergence occurs with respect to species. On evolutionary time scales, lineages of organisms acquire novel traits via evolutionary mechanisms — most commonly the mechanism of natural selection.

Diachronic emergence of properties is not necessarily limited to the biological. Many physical and social entities or systems acquire novel properties through the diachronic operation of mechanistic processes — mountain ranges, atmospheres, stars, galaxies, political parties, economic markets, and nation-states, to name just a few. Of course, the mere fact that some entity acquires properties gradually via a mechanistic process is not sufficient to classify it as emergent, but these examples at least suggest the possibility of the diachronic emergence of properties.

One last category of emergent that is important in many scientific discussions of emergence is the emergence of patterns (Winning & Bechtel, 2019). Patterns are of a higher order than entities, activities or properties, because when a pattern emerges, it can be a pattern of any of these things. For instance, there can be a pattern in distribution of any of these things. Take for example oscillatory patterns. When pendulums oscillate, what oscillates is the position and momentum of the weight; when circuits oscillate, what oscillates is current; and when populations oscillate (say due to procreation and predation), what oscillates is the size or density of a population within its environment; when markets oscillate, what oscillates are supply, demand and price. Other examples of emergent patterns include patterns of motion in flocks of birds or schools of fish, or, to consider a purely digital example, patterns in the emergence, disappearance or motion of shapes within cellular automata, exemplified by Conway's game of life.

Because it is higher order, the tools for studying pattern emergence are abstract and mathematical, including dynamical systems theory, chaos theory, cellular automata, and network analysis. These mathematical theories provide domain-independent tools for describing patterns in and dynamics of systems, processes, events, etc. They are especially suited to describing cases of diachronic emergence of patterns, like transitions between stable and chaotic behavior, or phase transitions

in states of matter. From a mechanistic point of view, patterns emerge because of how mechanisms are organized, and these mathematical tools allow one to describe these patterns in an abstract and general way.⁶

11.4 Autonomy, Holism, and Novelty in Mechanistic Emergence

In the previous section I've argued that the dependence required for emergence should be understood as mechanism-dependence. In this section I turn to the other three not wholly independent criteria – autonomy, holism and novelty. My strategy will be to run through a series of suggestions about what features might be required in order for a system to meet these criteria, and show how they can be fruitfully explicated within the mechanistic framework.

11.4.1 *Non-aggregativity*

Consider first Wimsatt's suggestion (1997, 2000, 2007) that emergence occurs in systems whenever they are **non-aggregative**. A property of a system is non-aggregative to the extent that its existence or value depends upon how the parts of the system are arranged. Wimsatt interprets aggregativity as a kind of invariance or stability of properties under various kinds of manipulations or transformations of a system's parts. He identifies conditions required for aggregativity:

1. *IS (InterSubstitution)*: Invariance of the system property under operations rearranging the parts in the system or interchanging any number of parts with a corresponding numbers of parts from a relevant equivalence class of parts ...
2. *QS (Size Scaling)*: Qualitative similarity of the system property (identity, or if a quantitative property, differing only in value) under addition or subtraction of parts....
3. *RA (Decomposition and ReAggregation)*: Invariance of the system property under operations involving decomposition and reaggregation of parts
4. *CI (Linearity)*: There are no Cooperative or Inhibitory interactions among the parts of the system that affect this property. (Wimsatt, 2007, 280–281)

⁶Mechanistic approaches are sometimes seen as at odds with more abstract and mathematical approaches to pattern emergence, especially with regard to explanation. Explanation of emergent phenomena is often seen to depend upon getting away from mechanistic detail (O'Malley & Dupré, 2005; Silberstein & Chemero, 2013; Batterman & Rice, 2014), and disciplines like systems biology and systems neuroscience are seen as alternatives to mechanistic approaches. While I cannot argue the case here, it seems to me that this tension is much overblown, and in fact that tools such as dynamical systems theory are an important part of the toolkit for describing mechanisms of emergence. For further discussion see (Kaplan & Bechtel, 2011; Craver & Kaplan, 2018; Brigandt et al., 2018)

Any failure to meet these conditions would imply emergence, with different failures yielding different varieties of emergence. On this view, emergence is ubiquitous, and non-emergent properties are the exception rather than the rule, since most properties of composite systems depend not just upon what parts the composite has, but how those parts are arranged.

A homely example should make clear why non-aggregativity is the usual case. Lawn mowers have many properties, but perhaps the most salient is that lawnmowers can cut laws. But the ability of the lawnmower to cut lawns depends not just upon the parts which make up the lawnmower, but upon those parts being assembled in the right way. Lawn-mowing is an activity which the lawn mower can engage in but which its components cannot. Moreover, you cannot typically take away parts of a lawnmower in a way that gradually degrades its lawn-mowing capacity.

Although Wimsatt claims all non-aggregative properties are emergent, it isn't clear that non-aggregativity alone guarantees that a system will meet all of the four conditions identified at the outset. Clearly it meets the dependence criterion, since emergent properties mechanistically depend upon the components of systems and mechanisms. Equally clearly, non-aggregativity is a kind of holism. If "aggregation" is analogous to summing, non-aggregativity entails that the whole is more than the sum of its parts.

Novelty, however, does not seem guaranteed by non-aggregativity. It is true that organized mechanical systems like the lawn mower will have properties and abilities that their parts do not – only the whole lawnmower can mow the lawn – but some failures of aggregativity don't yield genuinely new properties. For instance, in an electrical circuit, changing arrangements of resistors from serial to parallel changes the amount of current flowing, but it won't introduce a new kind of property. Most importantly, there is no obvious link between non-aggregativity and autonomy. The fact that properties of a system are organization-dependent does not by itself seem to imply any sense in which the system is autonomous from the components upon which it depends.

We may conclude I think that emergent properties will necessarily be non-aggregative, but that non-aggregativity is not a sufficient condition for emergence. Alternatively one might take non-aggregativity to be a weak form of emergence

11.4.2 *Externalism*

Non-aggregativity implies a weak more-than-the-sum-of-its-parts kind of holism, but it certainly does not block reductive explanation. Perhaps other sources of holism will yield stronger varieties of emergence. One possible source much discussed in philosophy of mind and cognitive science is externalism. Externalism is generally taken to come in two varieties. Passive externalism is a view about mental content, exemplified by Putnam's (1975) motto that "meanings ain't in the head." More recent discussions have focused on "active externalism" — sometimes called theories of 4E (embodied, embedded, extended, enacted) cognition (Newen et al., 2018).

Active externalist theories point to the ways that cognitive agents solve problems or complete tasks using resources that extend beyond their brain, and often beyond their body. Examples include the ways humans use fingers or pencil and paper or calculators to help solve math problems, the way that our bodies rely on environmental affordances to simplify search and movement tasks, and the ways that perception of objects requires movement about and interaction with those objects. The common theme of externalist approaches to cognition is that they suggest that an account of cognitive processes cannot be given which localizes cognition within the “naked brain.” Interactions with entities outside of the brain is required for an agent to acquire or exhibit a cognitive capacity. This is a distinctive kind of holism, since we find a property (here a cognitive capacity) that we typically ascribe to a cognitive agent in fact depends upon the agent acting within an encompassing environment. 4E cognition is amenable to mechanistic explanation (Miłkowski et al., 2018), but the mechanisms are necessarily wider than those working within the brain alone. These emergent capacities are also novel in the straightforward sense that they only emerge as the brain interacts with its environment.

While active externalism is a cognitive phenomenon, it seems to be an instance of something that occurs in many systems, a phenomenon we might call emergence as non-locality. This kind of emergence occurs whenever some activity or capacity that is often attributed to an entity in fact depends for its existence upon features of or interactions with the world beyond the entity’s boundaries. For instance, if it is in fact that case that scientific knowledge is essentially social in character, it follows that this kind of knowledge can only arise through organized interactions of scientific communities, rather than being localized in individual scientists. Similarly, in ecology, resilience is a property that belongs to an ecosystem rather than its parts. While individual organisms or species may be resilient in certain respects, the ecosystem’s ability to recover from shocks and stresses cannot be reduced to the resilience of its parts.

11.4.3 Downward Causation

Another feature widely held to characterize systems with emergent properties is downward causation. Many common sense and scientific causal claims appear to express relations in which higher level activities, events or properties causally influence happenings at a lower level. Mental causation seems to have this character. Undertaking a meditative exercise can have effects on physiological processes like heart beats. More generally, any time a perception or decision is followed by bodily action, as when my fingers press the keyboard as I try to explain downward causation, mental events seem to produce physical changes in the body. Moreover, downward causation is by no means limited to the mental. In fluid dynamics, large scale convection currents seem to causally constrain elements within those currents (Bishop & Silberstein, 2019). Social facts or events seem to exert downward causation on individual persons (Sawyer, 2004). In ecology and evolution, system level properties like population density affect individual fitness (Millstein, 2006).

A significant source of the recent resurgence of interest in emergent phenomena has been the need to make sense of these kinds of causal relations. Much of it has been motivated by Kim's causal exclusion argument, which was developed as an objection to non-reductive physicalism (Kim, 1993). Kim claims that if mental or other higher-level properties supervene on physical properties, the mental or higher-level properties cannot have causal efficacy, since the physical properties upon which they supervene are already sufficient for their effects.

There is now a substantial philosophical literature which attempts to elucidate downward causation as it occurs in disciplines across the physical, life and social sciences, with much of it aiming to show that the phenomenon is ubiquitous, non-mysterious and explicable by looking at the structure of mechanisms (Ellis et al., 2012; Paolini Paoletti & Orilia, 2017). Craver and Bechtel (2007), for instance, argue that top-down causation is actually a hybrid between interlevel mechanistic constitution relations and intralevel causal relations which they call "mechanistically mediated effects." Others argue that in certain kinds of mechanisms, higher level system or process variables are the causally relevant or difference making features, and form the basis for explanation and intervention (Glennan, 2010; Woodward, 2021). Relatedly, others have tried to understand the notion of higher level cause in terms of concepts of constraint and control (Kistler, 2009; Bechtel, 2017).

While there are disagreements about the proper metaphysical interpretation of top-down causation, there is a broad consensus that top-down causation is a widespread phenomenon, and that its occurrence is explained by the mechanistic structure of systems and processes in which it occurs. Different varieties of top-down causation may arise as the result of different kinds of mechanisms (Ellis, 2011). However exactly we explicate it, downward causation seems to be an important mark of the emergent. Top-down causation implies a kind of holism (system level properties cause, control or constrain activities of the parts) and a kind of autonomy (system level properties are the properties that make a difference). Moreover, systems or processes that exhibit top-down causal influence acquire properties and capacities that are novel relative to the properties and activities of their constituents.

11.4.4 *Self-Organization*

Another commonly cited characteristic of emergent systems and processes is **self-organization**. The concept of self-organization has a long history (Keller, 2008) and is not easily defined, but the general feature of self-organizing systems (entities) and processes (activities) is that they form and maintain themselves in the absence of external control. These processes of formation and maintenance arise spontaneously out of the local activities and interactions of components of these systems and processes. Examples of self-organization occur across a range of domains. Some examples include processes of crystal formation, processes of membrane formation

in cells, the processes that direct collective movements of herds of animals or flocks of birds, and the economic processes that lead to the formulation of markets and the establishment of prices.

A precondition for systems and processes to self-organize is a feature of mechanistic organization I call “affinitive organization” (Glennan, 2017). An interaction is affinitive to the extent that it is directed by the dispositions of the interactors rather than the direction or arrangement of an external controller. Consider as an example the difference between a cellular membrane and a brick wall. The membrane self-organizes because of its component entities, phospholipid molecules have hydrophilic heads and hydrophobic tails, which will in an aqueous environment spontaneously aggregate into sheets. Bricks on the other hand are not moved by attractive and repulsive forces to line up; you need a bricklayer to line them up and cement them together. It is precisely this absence of a controller that yields the sense in which self-organizing systems and processes are autonomous.

A related kind of organization is **self-maintenance**. A self-maintaining system is one that carries with it the capacity to maintain its properties and functions, repair damage, and so forth. Organisms are paradigms of self-maintaining systems; they have the capacity to maintain their state (e.g., concentrations of metabolites, temperature), to repair damaged tissues, and to destroy infectious agents. In contrast to a car, you do not normally have to take your body to the shop to repair a scratch.

11.4.5 Multiple Realization and Dynamical Autonomy

Multiple realization arguments have been employed to defend non-reductive physicalism, and, to the extent that that emergence is understood to be opposed to reduction, multiply realizable properties are natural candidates for emergents. Realization is a dependence relation, and it is natural to interpret it as a species of mechanism-dependence. Consider a well-known toy example, the capacity of certain tools to remove corks from wine bottles (Shapiro, 2000). This capacity is realized by different kinds of mechanisms in different kinds of corkscrews. For instance, a waiter’s corkscrew operates by twisting a screw into the center of the cork, swinging a hook attached to the device’s handle onto the bottle’s lip, and pulling up on the handle, which uses the hook as the fulcrum of a lever to pull out the cork. A winged corkscrew by contrast has a body with a collar that can rest on the lip of the bottle, and a screw mounted in the center of the body that can move through the collar as it is twisted into the cork. When the screw is inserted, two levers attached to the screw by gears (the wings) are driven up, and pushing down on the wings pulls the cork out through the collar.

Multiple realization allows for token reductive explanations, since each token system (like a corkscrew) has a token mechanism that gives it that capacity. Multiple realization arguments are instead focused on types. A given type of capacity, like the capacity to remove corkscrews, can be realized by many different types of mechanisms. Historically the point of multiple realizability arguments was to argue for the

explanatory autonomy of special sciences (Fodor, 1974). Explanations in economics, for instance, can formulate principles about the relationship between supply, demand and price, without concerning itself with the physical realizations of money or the mechanisms by which money and goods trade hands.⁷

Plausibly, multiple realizability is enough to yield a kind of autonomy sufficient for emergence. While any token of the emergent capacity will depend upon a particular realizing mechanism, different tokens of this capacity can and often do depend upon different types of mechanisms. This means the capacity as such does not depend upon a particular mechanism, and is hence autonomous from it.

What Wimsatt calls dynamical autonomy is a conceptual cousin of multiple realizability, but while multiple realization focuses on how different tokens of a particular property or kind can be realized by distinct mechanisms, dynamical autonomy points instead to the ways in which a single token macro-state or property of a system can persist even in the face of changes to that token's micro-state and mechanisms. As Wimsatt puts it "dynamical autonomy ...entails that most ... micro-level changes don't make a causal difference at the macro-level (Wimsatt, 2007, 218).

Examples of dynamical autonomy abound. For instance, the states of an organism can remain stable at the macro level even as micro-level changes like the birth and death of cells are constantly occurring. Similarly, psychological states of human being can persist even as many neurological features of the person are in flux. It is this persistence that guarantees that it is the macro-states rather than the micro-states which are causally relevant to a system's behavior. Dynamical autonomy seems to capture the both the dependence and autonomy required for emergence. There is dependence, because the macro-state needs a micro-state to realize it, but there is autonomy, because the macro-state and its causal powers can persist in the face of changes to the micro-state.

11.4.6 *Transformation and Fusion*

Paul Humphreys argues that philosophical accounts of emergence have too often ignored diachronic emergence, and offers as a remedy an account of what he calls transformational emergence (2016, sec. 2.1.1). Humphreys motivates the account by considering the emergent behavior of mobs. What we observe in so-called "mob psychology" is that the behavior of individuals within the mob is transformed, so that those whose normal dispositions might be friendly and non-violent exhibit a

⁷Multiple realizability arguments have not aged that well, because it is increasingly apparent that the implementing mechanism makes a difference to the realized capacity (Polger & Shapiro, 2016). Not all corkscrew mechanisms are equally good at removing corks, and all have their quirks. Or, more seriously, it is evident that changes to the medium of financial transactions can profoundly affect the market's behavior. But even if multiple realizable properties are harder to come by than was once thought, it may be still that such properties are good candidates for emergents.

new and destructive set of dispositions when they become part of a mob. As Humphreys describes the case, what emerges is a transformed individual behaving in accordance with a new set of generalizations – laws of abnormal psychology rather than of ordinary psychology. After such a transition, the transformed individuals act in ways that are quite different from their past selves.

Humphreys goes to some length to distinguish the transformation that occurs individuals in a mob from the change in behavior that one sees in ordinary crowds or in flocks of birds. At first sight they seem similar, since crowds and flocks, just like mobs, make their members behave in ways they would not on their own. But Humphreys claims that only in the case of the mob is there genuine ontological emergence. The reason he thinks is that in ordinary crowds or flocks nothing essential has changed in the dispositions of the individuals, while in the mob something has. Consideration of our own experience suggests that this distinction is important. If I am moved along in a crowd as I leave the stadium, my motion is severely constrained, so I have lost my individual agency, while, in contrast, were I caught up in the feelings of the mob, I might want to engage in the violent acts that others were engaging in; not just my actions but my beliefs and desires would be transformed.

Despite its initial plausibility, Humphreys' distinction between essential and accidental transformations is hard to sustain. The account presupposes that the individuals that form some composite have a set of essential and intrinsic characters that, together with their arrangement, fix the properties of the composite of which they are part. But this kind of essentialism is something we generally have reasons to reject. Whenever a composite is formed, it constrains and alters the components' activities, and it is extremely difficult to make a principled case that some but not all such transformations represent the transformation of the component itself. All we can say with certainty is that the component's behavior is transformed by its placement in the composite.⁸

Humphreys has identified a particularly strong form of transformation he calls fusion. He considers covalent bonding to be a paradigm case. Consider a hydrogen molecule consisting of two hydrogen atoms. Humphreys writes that when the atoms are close together, their individual identities disappear:

Although the standard treatments usually talk of two indistinguishable electrons, what one really has is a joint probability distribution within which there is no sense to be made of separate, coexisting particles. This means that if the original, spatially separate hydrogen atom 1 is identified with proton 1 and electron 1, and the original, spatially separate hydrogen atom 2 is identified with proton 2 and electron 2, within the bonded molecule it is no longer possible to say that hydrogen atom 1 (or 2) exists as an identifiable subunit. (Humphreys, 2016, p. 83).

⁸Although Humphreys (2016) uses mob psychology to introduce the notion of transformational emergence, he ends up doubting that it is a genuine case. His rationale is that an improved neuropsychological theory may show that the essential properties of individuals do not change in mobs, but only that they only manifest themselves in novel ways in mob-like environments. Given my skepticism that there are any essential psychological properties (and with them laws), I would argue that mob psychology is a good example of transformational emergence.

If this description is right, it does suggest that in this kind of bond, the identity of the components is lost in the fused entity. What emerges is a new entity — not just an arrangement of existing entities. Humphreys sees this a special case of transformation, where the individuals in the composite are not merely changed in their essential properties, but disappear into a new entity. Such processes, Humphreys plausibly believes, exemplify a strong sense of novelty and holism.

It is not obvious to me though that there is a principled and domain-neutral account as to when acts of composition lead to the disappearance of the components. Humphreys for instance argues (84–85) that when one kneads together two lumps of clay, the original lumps may not be lost, because at a molecular level the original lumps (now spatially distributed) could in principle be recovered. In contrast, speculatively, he suggests (86) that the unity government formed in the United Kingdom during World War II was an act of fusion, in the sense that the Labor and Conservative parties “disappeared” and were replaced by a new party with novel properties. My intuition is that properties of MPs might be more recoverable than properties of clay lumps, and that the properties of the unity government, as well as transformation of its members might be mechanistically explained. But whatever one’s intuitions, it seems that absent a firmer account of essential properties, all we will know is that in these transformational processes, both the components and the composites will change in dispositions and activities so as to exhibit the properties of novelty, autonomy and holism.

Humphreys’ examples of diachronic transformations are drawn primarily from the physical sciences, but the model is intended to be general, so it is worth mentioning a few much-discussed cases from other domains to which Humphreys’ account might be applied. One important case is intentionality. Just what intentionality is and where it comes from is of course a matter of controversy, but a prominent approach that seems to involve diachronic transformation is Dretske’s (2021) account of a “recipe for thought.” Dretske holds that a system acquires thoughts and other intentional states as it interacts with features of the environment which are relevant to its needs or interests. The system starts with some primitive capacities, but it only acquires genuine intentional states when environmental processes trigger these capacities. If the system has the capacity to rewire its responses based upon this input, it should over time begin to acquire something like genuine representations (and misrepresentations) — or so the theory goes.

Dretske’s recipe for thought is meant to account for how thought and representation develop in natural systems, but we can tell a similar story for artificial systems that are capable of being trained. The deep neural networks used in speech or facial recognition technologies are systems that use their environment (a set of training data) to tune the network to recognize or classify words or faces. Similar techniques are used by chat bots and game-playing AIs that learn from their experience as they interact with you. While it seems improbable that these kinds of systems yet have the kinds of goals and interests required to acquire “genuine” intentionality, they certainly acquire novel capabilities through their interactions with the environment. The reason that it is so natural to call these sorts of phenomena emergent is that one can’t really build in the capacity from the start. As Dretske emphases, you can’t just

add such a capacity “the way you add spices in a recipe for lasagna. Adding the function is more like *waiting* for the dough to rise” (2021, 357).

One last example worth mentioning is what Paul (2014) has called “transformative experiences.” Paul argues that humans cannot make rational decisions (in the sense of decision theory) regarding experiences – like having a child or undergoing a religious conversion– which will be both epistemically and personally transformative. Personally transformative experiences can make changes to one’s basic psychology – to one’s likes, values and personality – such that one emerges as a qualitatively different person. Following Humphreys’ model of transformation, such personal transformations, when they occur, would be cases of diachronic emergence. Paul’s account may run into the same trouble that Humphreys had with distinguishing genuinely transformative changes from mere changes in behavior, but, to the extent that such a distinction can be maintained, this seems an interesting case of diachronic emergence of new persons and properties.

11.5 Conclusion: But is this Really Emergence?

I hope in this paper to have shown that the opposition between mechanism and emergence is based on a misunderstanding, and that core features of emergent phenomena – dependence, autonomy, holism and novelty can be explicated in mechanistic terms. In addition, the mechanistic turn can shed light on differences in varieties of emergence. The distinction between mechanisms that produce vs mechanisms that underlie provides an analysis of the distinction between diachronic and synchronic emergence, and various interpretations of novelty, holism and autonomy can be shown to arise from different kinds of mechanistic organization.

I expect though that my proposed rapprochement between mechanism and emergence will be met with skepticism. One reason is that the mechanistic conception of emergence makes emergent properties the rule rather than the exception. It violates what Humphreys has called the rarity heuristic, which holds that any account “that makes ontological emergence commonplace has misidentified the criteria for emergence” (Humphreys, 2016, 54). But, as Humphreys effectively argues, there’s no clear argument for this heuristic, and we find ample evidence across the sciences for phenomena that can be characterized as both dependent upon and distinct from some base from which they emerge.

But I expect a deeper source of skepticism may lie in the way in which philosophers have understood the historical relationship between concepts of mechanism and emergence. Aristotle is often credited with being the first emergentist, because of his metaphysics of matter and form. Substances depend for their existence upon matter, but it is the way that this matter exemplifies form that gives a substance its novel properties and causal powers. But, the story goes, Aristotelian metaphysics was rejected in the scientific revolution and replaced by mechanical philosophy, an austere metaphysics in which all things in the world – at least all things in the material world – were nothing but matter and motion. Since that time, various

incarnations of this reductionist nothing-but-ism have thrived, from Laplacian determinism and de La Mettrie's "L'Homme Machine, to Wittgenstein's and Russell's logical atomism, to many modern articulations of microphysicalism in contemporary metaphysics. Humphreys calls the general strategy found in these sources **generative atomism** – because it assumes that there is some fundamental level of basic immutable entities, and that all higher levels of things are generated by dynamical and compositional principles or laws from these fundamental things. When Broad drew the distinction between mechanism and emergence by arguing that emergent phenomena cannot "be deduced from the most complete knowledge of the behavior of its components, taken separately or in other combinations, and of their proportions and arrangements in this whole" (1925, 59), it is clear that he was assuming an austere nothing-but form of mechanism.

Contemporary philosophical research on mechanism though has rejected this austere approach. The new mechanistic account is grounded in the practices of the life and social sciences. Those practices do not seek out a set of privileged atoms and properties from which all else is generated, but look instead at phenomena within a domain, and seek to identify particular kinds of entities, activities and interactions, and show how they are organized so as to be causally and constitutively responsible for their phenomena. As Bechtel and Richardson (2010, xliv) put it, some kinds of mechanisms exhibit emergent behaviors that are "neither weak nor epistemic." Mechanistic phenomena are not simply generated by the activities of a set of immutable parts. They instead can (as discussed above) depend upon varieties of feedback and top-down causation (or something like it) whereby the whole influences the part.⁹ Investigation of these processes is, as Wimsatt has long argued, reductive, but not eliminative, and is piecemeal and local. The resultant ontology is, as he puts it, a rainforest ontology – with intricate dependencies, but also rich with novelty.¹⁰

The skeptic might still question how much distance mechanists can put between themselves and generative atomism. Given that novel entities and activities are going to be mechanism-dependent, just how novel can they really be? One way to pose this question is by appealing to the commonly held distinction between weak and strong emergence. As Jessica Wilson (2016, 2021) sees it, weak and strong emergence are both genuine ontological emergence, but represent different alternatives to traditional physicalism. Weak emergence follows the path of non-reductive physicalism, most commonly by appealing to the multiple realizability of

⁹Bechtel and Richardson's account is offered in the context of systems biology, but the conditions they require are generalizable to other domains. They argue that there are two necessary conditions for emergence. First, (some) interactions between components must be non-linear, and second, the state of the whole system must be characterized in terms of state-independent properties. These properties, which include properties of the whole system and its environment determine the properties of the components. While I haven't formulated conditions for emergence in just this way, Bechtel and Richardson's conditions seem to imply top-down causation and transformation, which are both inimical to generative atomism.

¹⁰These themes, which appear in much of Wimsatt's work since the 1970s, are nicely summarized in the introduction to (Wimsatt, 2007).

higher-level properties to justify claims of their autonomy. But on Wilson's view, weak emergence does not introduce genuinely novel causal powers. For that, one needs something like new fundamental laws to characterize the powers of the emergents.

There are, as I see it, three possible responses to such skepticism. First, one might grant that the kind of emergence compatible with mechanism is indeed weak emergence, but allow that there may be some rare phenomena – perhaps consciousness or agency – which exhibit strong emergence. A second approach is to argue that the strong emergentist position ultimately collapses into weak emergentism. The best and only kind of emergence we will find fails to introduce genuinely novel causal powers. The third and perhaps best response might be to question the basis for the distinction itself. The most common explications of strong emergence are cashed out in terms of levels, domains and laws. Novel causal powers will be expressed by causal laws characterizing relations between the novel entities' properties. But there are many strands of recent philosophy of science that suggest that analysis of laws and causes may not be the right way to go. Perhaps there are very few laws, or perhaps, as I prefer (Glennan, 2017), we should think of laws simply as descriptions of the behavior of mechanisms. If we think about laws and causes in this way, we may be forced to a different view of what would count as novelty, and mechanism might just provide novelty enough.

I have certainly not said enough here to convince skeptics that all emergent phenomena are within the reach of mechanism – and I am far from certain myself. But I hope I have said enough to show that the supposed incompatibility of mechanism and emergence reflects a misunderstanding of what mechanisms are. Mechanisms don't rid us of emergent phenomena; they show us how they work.

References

- Batterman, R. W., & Rice, C. C. (2014). Minimal model explanations. *Philosophy of Science*, 81, 349–376. <https://doi.org/10.1086/676677>
- Bechtel, W. (2017). Top-down causation in biology and neuroscience: Control hierarchies. In M. P. Paoletti & F. Orilia (Eds.), *Philosophical and scientific perspectives on downward causation* (pp. 203–224). Routledge. <https://doi.org/10.4324/9781315638577-12>
- Bechtel, W., & Richardson, R. (2010). *Discovering complexity: Decomposition and localization as strategies in scientific research*. Reissue. MIT Press.
- Bishop, R., & Silberstein, M. (2019). Complexity and feedback. In S. Gibb, R. Hendry, & T. Lancaster (Eds.), *The Routledge handbook of emergence*. Routledge.
- Brigandt, I., Green, S., & O'Malley, M. A. (2018). Systems biology and mechanistic explanation. In S. S. Glennan & P. M. K. Illari (Eds.), *The Routledge handbook of mechanisms and mechanical philosophy*. Routledge.
- Broad, C. D. (1925). *The mind and its place in nature. The mind and its place in nature*. Kegan Paul, Trench, Trubner & Ltd.. <https://doi.org/10.4324/9781315824147>
- Craver, C. F. (2007). *Explaining the brain*. Oxford University Press.
- Craver, C. F., Glennan, S. S., & Povich, M. (2021). Constitutive relevance & mutual manipulability revisited. *Synthese*, 199, 8807–8828. <https://doi.org/10.1007/s11229-021-03183-8>. Springer Netherlands.

- Craver, C. F., & Bechtel, W. (2007). Top-down causation without top-down causes. *Biology and Philosophy*, 22, 547–563. <https://doi.org/10.1007/s10539-006-9028-8>
- Craver, C. F., & Kaplan, D. M. (2018). Are more details better? On the norms of completeness for mechanistic explanations. *The British Journal for the Philosophy of Science*, 0, 1–33. <https://doi.org/10.1093/bjps/axy015>
- Craver, C. F., & Tabery, J. G. (2016). Mechanisms in science. In E. N. Zalta (Ed.), *Stanford encyclopedia of philosophy*, Fall 2016.
- Dretske, F. (2021). A recipe for thought. In D. J. Chalmers (Ed.), *Philosophy of mind: Classical and contemporary readings* (2nd ed., pp. 351–358). Oxford University Press.
- Ellis, G. F. R. (2011). Top-down causation and emergence: Some comments on mechanisms. *Interface Focus*, 2, 126–140. <https://doi.org/10.1098/rsfs.2011.0062>
- Ellis, G. F. R., Noble, D., & O'Connor, T. (2012). Top-down causation: An integrating theme within and across the sciences? *Interface Focus*, 2, 1–3. <https://doi.org/10.1098/RSFS.2011.0110>. Royal Society.
- Fodor, J. (1974). Special sciences (or: The disunity of science as a working hypothesis). *Synthese*, 28, 97–115.
- Gibb, S., Hendry, R. F., & Lancaster, T. (2019). *The Routledge handbook of emergence*. Routledge. <https://doi.org/10.4324/9781315675213>
- Glennan, S. S. (2010). Mechanisms, causes, and the layered model of the world. *Philosophy and Phenomenological Research*, 81, 362–381. <https://doi.org/10.1111/j.1933-1592.2010.00375.x>
- Glennan, S. S. (2017). *The new mechanical philosophy*. Oxford University Press.
- Glennan, S. S. (2021). Corporeal composition. *Synthese*, 198, 11439–11462. <https://doi.org/10.1007/s11229-020-02805-x>. Springer Netherlands.
- Glennan, S. S., & Illari, P. (2018a). Introduction: Mechanisms and mechanical philosophies. In S. S. Glennan & P. Illari (Eds.), *The Routledge handbook of mechanisms and mechanical philosophy* (pp. 1–10). Routledge. <https://doi.org/10.4324/9781315731544>
- Glennan, S. S., & Illari, P. (2018b). Varieties of mechanisms. In *The Routledge handbook of mechanisms and mechanical philosophy*. <https://doi.org/10.4324/9781315731544>
- Glennan, S. S., Illari, P., & Weber, E. (2021). Six theses on mechanisms and mechanistic science. *Journal for General Philosophy of Science*. Springer Netherlands. <https://doi.org/10.1007/s10838-021-09587-x>
- Humphreys, P. (2016). *Emergence: A philosophical account*. Oxford University Press. <https://doi.org/10.1093/acprof>
- Kaplan, D. M., & Bechtel, W. (2011). Dynamical models: An alternative or complement to mechanistic explanations? *Topics in Cognitive Science*, 3, 438–444.
- Keller, E. F. (2008). Organisms, machines, and thunderstorms: A history of self-organization, part one. *Historical Studies in the Natural Sciences*, 38, 45–75. <https://doi.org/10.1525/HSNS.2008.38.1.45>
- Kim, J. (1993). The nonreductivist's trouble with mental causation. In *Supervenience and mind: Selected philosophical essays*. Cambridge University Press.
- Kistler, M. (2009). Mechanisms and downward causation. *Philosophical Psychology*, 6, 595–609.
- Krickel, B. (2018). *The mechanical world: The metaphysical commitments of the new mechanistic approach*. Springer.
- Machamer, P., Darden, L., & Craver, C. F. (2000). Thinking about mechanisms. *Philosophy of Science*, 67, 1–25.
- Miłkowski, M., Clowes, R., Rucińska, Z., Przeglasińska, A., Zawadzki, T., Krueger, J., Gies, A., et al. (2018). From wide cognition to mechanisms: A silent revolution. *Frontiers in Psychology*, 9. <https://doi.org/10.3389/fpsyg.2018.02393>
- Millstein, R. L. (2006). Natural selection as a population-level causal process. *The British Journal for the Philosophy of ...*, 57, 627–653.
- Newen, A., de Bruin, L., & Gallagher, S. (2018). *The Oxford handbook of 4E cognition*. Oxford University Press.

- O'Connor, T. (2021). Emergent properties. In E.N. Zalta (Ed.), *Stanford encyclopedia of philosophy*. Winter 2021.
- O'Malley, M. A., & Dupré, J. (2005). Fundamental issues in systems biology. *BioEssays*, 27, 1270–1276. <https://doi.org/10.1002/bies.20323>
- Paolini Paoletti, M., & Orilia, F. (2017). *Philosophical and scientific perspectives on downward causation*. Routledge.
- Paul, L. A. (2014). *Transformative experience*. Oxford University Press.
- Polger, T. W., & Shapiro, L. A. (2016). *Multiple realization book*. Oxford University Press.
- Putnam, H. (1975). The meaning of “Meaning.”. In *Minnesota studies in the philosophy of science philosophy of science. Vol. VII. Language, mind and knowledge* (pp. 131–193). University of Minnesota Press. <https://doi.org/10.1515/9783110871241.110>
- Sawyer, R. K. (2004). The mechanisms of emergence. *Philosophy of the Social Sciences*, 34, 260–282.
- Shapiro, L. A. (2000). Multiple realizations. *The Journal of Philosophy*, 97, 635–654.
- Silberstein, M., & Chemero, A. (2013). Constraints on localization and decomposition as explanatory strategies in the biological sciences. *Philosophy of Science*, 80, 958–970. <https://doi.org/10.1086/674533>
- Wilson, J. M. (2016). Metaphysical emergence: Weak and strong. In T. Bigaj & C. Wüthrich (Eds.), *Metaphysics in contemporary physics*. Brill.
- Wilson, J. M. (2021). *Metaphysical emergence. Metaphysical emergence*. Oxford University Press. <https://doi.org/10.1093/oso/9780198823742.001.0001>
- Wimsatt, W. C. (1997). Aggregativity: Reductive heuristics for finding emergence. *Philosophy of Science*, 64, S372–S384.
- Wimsatt, W. C. (2000). Emergence as non-Aggregativity and the biases of reductionisms. *Foundations of Science*, 5, 269–297.
- Wimsatt, W. C. (2007). *Re-engineering philosophy for limited beings*. Harvard University Press.
- Winning, J., & Bechtel, W. (2019). Being emergence vs. pattern emergence: Complexity, control and goal-directedness in biological systems. In S. Gibb, R. Hendry, & T. Lancaster (Eds.), *The Routledge handbook of emergence* (pp. 134–144). Routledge.
- Woodward, J. (2021). Downward causation defended. In *Top-down causation and emergence* (Vol. 439, pp. 217–251). Springer. https://doi.org/10.1007/978-3-030-71899-2_9

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